DBT PORTFOLIO PROJECT

LAYER: MART (FINAL OUTPUT)

DBT Mart Model

Mart | Tableau | Wide Table | LEFT JOIN | Analytics | Snowflake

Project Summary

The final wide mart table that joins all intermediate statistical models into one flat row per entity/date -- moving averages, simple linear regression fitted values and residuals, Value at Risk at three confidence levels, and Sharpe Ratio. This is the table Tableau connects to directly. ARIMA, GARCH, MLR, and Exponential Smoothing joins are coded and ready to activate.

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1 PROJECT OVERVIEW

Model Name:	fct_claims_analytics
Layer:	Mart (final output layer)
Materialization:	table (Tableau connects directly)
Grain:	One row per entity_id + period_date
Active Joins:	int_moving_average, int_simple_linear_regression, int_value_at_risk, int_sharpe_ratio
Pending Joins:	int_arima, int_garch, int_multiple_linear_regression, int_exponential_smoothing
Audience:	Tableau dashboards, Power BI reports, Finance leadership, Operations

2 BUSINESS PROBLEM

Tableau cannot join nine separate models on the fly without performance degradation. Finance and operations teams needed a single, pre-joined table they could connect to directly -- one row per entity and period containing every statistical metric side-by-side. This mart eliminates the need for Tableau data blending or complex calculated fields, putting all analytical columns in one place.

- Single source of truth: All downstream dashboards point to one table -- no divergent metrics.
- Pre-aggregated joins: All intermediate models aggregated before joining -- no fan-out row multiplication.
- Moving averages (7 and 30 period) from int_moving_average -- primary time-series spine.
- SLR fitted values and residuals from int_simple_linear_regression -- trend line for Tableau.
- VaR at 90/95/99% from int_value_at_risk -- entity-level risk metrics repeated across dates.
- Sharpe Ratio (periodic and annualised) from int_sharpe_ratio -- performance benchmarking.
- Commented joins: ARIMA, GARCH, MLR, and Exponential Smoothing ready to activate when validated.

3 TECHNICAL EXPLANATION

- Spine: int_moving_average drives the row grain (entity_id + period_date) -- all other models join to it.
- LEFT JOIN on entity_id + period_date for int_simple_linear_regression -- preserves all spine rows even if SLR has gaps.
- LEFT JOIN on entity_id only for int_value_at_risk and int_sharpe_ratio -- these are entity-level aggregates, repeated across all dates for that entity.
- Commented-out sections: int_arima, int_garch, int_multiple_linear_regression, int_exponential_smoothing are written and ready -- remove comment markers to activate.
- Materialized as TABLE: Tableau requires fast query response -- pre-computation is essential.
- Uses {{ ref() }} exclusively -- all joins go through the dbt model graph, not raw table names.

Key Techniques: {{ ref() }} model references, LEFT JOIN spine pattern, entity vs date grain joins, commented activation pattern

4 FULL MODEL CODE

```
{{ config(materialized='table') }}

-- One wide row per entity + date, joining all statistical model outputs
-- This is what Tableau connects to directly

select
    ma.entity_id,
    ma.payer,
    ma.state,
    ma.specialty,
    ma.period_date,
    ma.metric_value,

    -- Moving averages (7-period and 30-period)
    ma.moving_avg_7,
    ma.moving_avg_30,

    -- Simple Linear Regression: fitted values and residuals
    lr.fitted_value     as slr_fitted_value,
    lr.residual         as slr_residual,
    lr.beta_0           as slr_intercept,
    lr.beta_1           as slr_slope,

    -- Value at Risk (entity-level, repeated across dates)
    var.var_90_pct,
    var.var_95_pct,
    var.var_99_pct,

    -- Sharpe Ratio (entity-level, repeated across dates)
    sr.sharpe_periodic,
    sr.sharpe_annualised

    -- Pending activation (remove comment markers when models are validated):
    -- ar.diff_1, ar.ar_lag_1, ar.ar_lag_2, ar.ar_lag_3,
    -- gr.conditional_variance, gr.conditional_volatility,
    -- mlr.beta_0, mlr.beta_1, mlr.beta_2, mlr.fitted_value, mlr.residual,
    -- es.es_smoothed_value_alpha_0_3

from {{ ref('int_moving_average') }} ma

left join {{ ref('int_simple_linear_regression') }} lr
    on ma.entity_id  = lr.entity_id
    and ma.period_date = lr.period_date

left join {{ ref('int_value_at_risk') }} var
    on ma.entity_id = var.entity_id

left join {{ ref('int_sharpe_ratio') }} sr
    on ma.entity_id = sr.entity_id

-- left join {{ ref('int_arima') }}                    ar  on ma.entity_id = ar.entity_id
-- left join {{ ref('int_garch') }}                    gr  on ma.entity_id = gr.entity_id
-- left join {{ ref('int_multiple_linear_regression')}} mlr on ma.entity_id = mlr.entity_id
```

```
-- left join {{ ref('int_exponential_smoothing') }}    es  on ma.entity_id = es.entity_id
```

5 EXAMPLE OUTPUT (SELECTED COLUMNS)

entity_id	period_date	metric_value	ma_7	slr_fitted	VaR95%	sharpe_a
E-001	2025-01-01	4250.75	4250.75	4118.67	-0.0051	5.347
E-001	2025-02-01	4380.20	4315.48	4137.12	-0.0051	5.347
E-001	2025-03-01	4520.80	4383.92	4155.57	-0.0051	5.347
E-002	2025-01-01	3180.10	3180.10	3063.10	-0.0204	2.136

** VaR and Sharpe are entity-level aggregates -- they repeat across all period_date rows for that entity.*